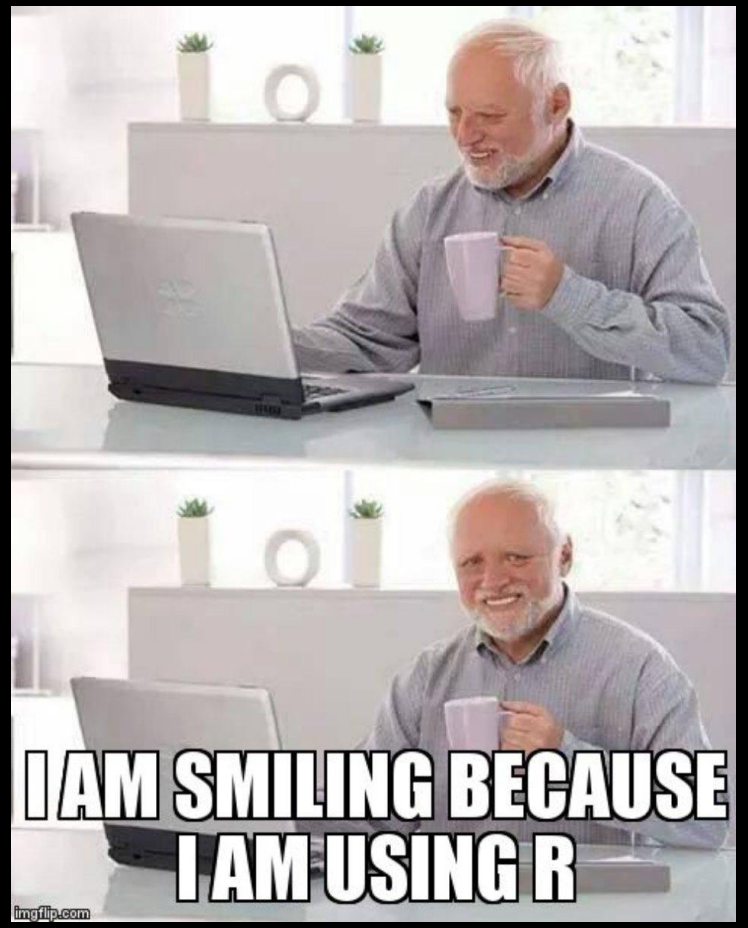


MES 503 Lab 9

Kayla Blincow, PhD

Announcements

- Lab Assignment 2 due tomorrow ☺
- Quiz on Wednesday!
  - Two-Way ANOVAs + Multiple Linear Regression

TENTATIVE SCHEDULE OF TOPICS – MES 503 Fall 2026

		Mondays (Lectures)		Wednesdays (Labs)	
Week	**Date**	**Lecture Topic**	**Date**	**Lab topic**	**Assessments**
1	8/17	Introduction to Statistics & Experimental Design I	8/19	Introduction to R and RStudio	
2	8/24	Introduction to Statistics & Experimental Design II	8/26	Intro to ggplot2	
3	8/31	Correlation	9/2	Correlation in R	Quiz 1
4	9/7	LABOR DAY (No Class)	9/9	Intro to Data wrangling	Lab 1 Due (9/10)
5	9/14	Statistical Hypothesis Testing & t tests	9/16	t Tests in R	
6	9/21	Linear Regression	9/23	Linear Regression in R	Quiz 2 (9/21) Lab 2 Due (9/24)
7	9/28	ANOVA	9/30	ANOVA in R	
8	10/5	ANOVA II (Two-Way & Nested) (virtual)	10/7	Two-Way & Nested ANOVA in R	Quiz 3
9	10/12	Multiple Linear Regression	10/14	Multiple Linear Regression in R	Lab 3 Due (10/15)
10	10/19	ANCOVA	10/21	ANCOVA in R	Quiz 4
11	10/26	Analysis of Frequencies	10/28	Analysis of Frequencies in R	Lab 4 Due (10/29)
12	11/2	Liberty Day – UVI Closed	11/4	Miscellaneous Fun R Topic	Quiz 5
13	11/9	Quantifying Variability/Figure and Data Sheet Design	11/11	Veterans Day – UVI Closed	
14	11/16	Reading for Stats Presentations	11/18	Reading for Stats Presentations	Lab 5 Due (11/17)
15	11/23	Applying this course to your thesis	11/25	Study Day	
16			12/3	**Final Exam**	



Sea Cucumbers!

- Sea cucumbers are overharvested across the globe
- They are highly susceptible to overfishing, because they are essentially defenseless to human exploitation and have density dependent reproduction
- Highly valuable in Asian markets
- One of the most highly trafficked animal products across US-Mexico border





Scenario

We are fisheries biologists that are interested in learning more about the sea cucumber fishery in Indonesia, the world's top producer of cukes. In particular, we are interested in testing how a suite of variables influence catch per unit effort in cuke fisheries. We survey cucumber fishers and get records of the CPUE across their main fishing grounds.



Our variables of interest are: depth, length of time fisher has been engaged in the fishery, fisher experience metric, region

Switch to ze code!

Dates and Times in R



Why dates/times are so gosh darned annoying…

- We have imperfect systems for tracking time
- Writing dates/times is often ambiguous which is hard for computers
- Yet, time is super important for almost all ecological questions!



How R handles dates/time.

- R will not automatically classify something as a date/time unless you tell it to
  - Thank goodness!
- Most common date/time classifying functions
  - as.Date() – for dates only
  - as.POSIXct() – for dates and times
  - lubridate package – tidyverse's answer to date/time issues



Symbol	Meaning	Example
%a	Abbreviated weekday name	Tue
%A	Full weekday name	Tuesday
%b	Abbreviated month name	Apr
%B	Full month name	April
%C	Century: the integer part of the year divided by 100	20
%d	Day of the month	09
%H	Hours as decimal number (00–23)	13
%I	Hours as decimal number (01–12)	1
%m	Month as number (01–12)	04
%M	Minute as number (00–59)	12
%p	AM/PM indicator for 12-hour time (%I)	PM
%S	Second as integer (00–61)	12
%u	Weekday as a decimal number (1–7, Monday is 1)	2
%w	Weekday as decimal number (0–6, Sunday is 0)	2
%y	2-Digit Year (00–99)	19
%Y	4-Digit Year	2019

Switch to code!

